TKV: Keep What the Basis Cannot Explain

A frozen low-rank KV cache, and the attention-free, query-agnostic selection criterion it provides

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Workshop

only one
configuration here with a worst-case guarantee: **TKV-R** (projection, not eviction)

0.93
and it ties the best method on Mistral-7B (H2O 0.67, StreamingLLM 0.60), 0.59× memory

0.77
TKV-R on TinyLlama: ties H2O, beats SnapKV (0.67) and StreamingLLM (0.60)

rank 8
all the criterion needs: one 8-dim projection of each pre-RoPE key

Compress Before the Question Arrives

Cached documents, multi-turn chat and shared prefixes compress the context **before** the question is known.

- **SnapKV / H2O** score tokens by how *past* queries attended to them, and need attention probabilities that fused kernels don't expose.
- With the question in view, SnapKV is near-exact (1.00). Without it, every method has to guess.

Can the keys alone tell us which tokens a future query will need?

TKV and Its Residual Criterion (TKV-S)

- **TKV**: fit a rank-r basis B to the **pre-RoPE** keys of the first 512 tokens, then freeze it (r=8 for scoring).
- Score every later token by its residual $\rho = \|k - B^\top Bk\| = \sqrt{\|k\|^2 - \|Bk\|^2}$.
- **TKV-S**: keep the top-M residuals exact (a min-heap while streaming). No attention scores, $\mathcal{O}(rd)$ per token.

A small basis captures what most tokens share; a token carrying *new* information is one it explains poorly.

Project, Don't Evict

RoPE is a rotation, so if a non-selected token is **kept as a projection** (TKV-R), then for any query q:

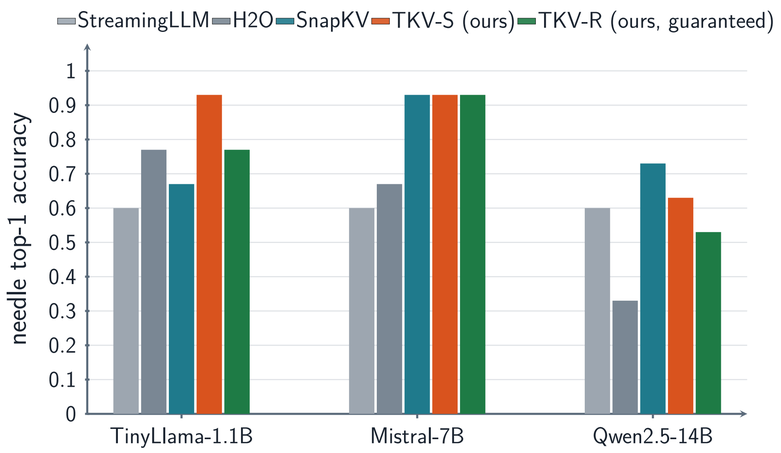
$$|q^\top \mathcal{R}_j k_j - q^\top \mathcal{R}_j \hat{k}_j| \le \|q\|\,\rho_j, \qquad \|p - \hat{p}\|_1 \le e^{2\varepsilon} - 1, \quad \varepsilon = \alpha \|q\| \max_{\text{not kept}} \rho_j.$$

Keeping the top-M residuals **minimizes this bound for all queries at once**.

But if tokens are **evicted**, no query-agnostic rule can have such a bound — ours or anyone's.

Drop a token and the error is set by the attention mass it would have received; a query aligned with an evicted key drives that mass to 1. So TKV-S, H2O, SnapKV and KVzip are all heuristics — judged empirically (right). **TKV-R is the member of the family with a guarantee — and it ties the best method on Mistral-7B (0.93) and ties H2O on TinyLlama (0.77), ahead of SnapKV and StreamingLLM.**

Result: Query-Agnostic Needle Retrieval



Matched KV memory (0.672× TinyLlama, 0.594× others), 30 prompts per bar (5 depths × 6 needles), context compressed before the question. TKV-R keeps non-selected tokens as projections, so it is the only bar with a worst-case guarantee.

TKV-S vs.	TinyLlama	Mistral-7B	Qwen-14B
StreamingLLM	p=.005	p=.005	tie
H2O	+5 (n.s.)	p=.02	p=.04
SnapKV	p=.02	tie	−3 (n.s.)

Fisher's exact test on correct answers out of 30.

H2O's cumulative attention favors early tokens and collapses on late needles (P = 0.07 / 0.03 at depth 0.9 on Mistral / Qwen). Query-aware SnapKV (seeing the question) scores 1.00: there's real headroom left.

Corrections Along the Way

- **Brand warmup.** Truncated updates are optimal only for the *retained state + new token*, not the full history. Valid online certificate: $\|E_t\|_F \le \|E_{t_c}\|_F + \Sigma_s\, \sigma_{R+1}(A_s)$.
- **Pre-RoPE bases**: post-RoPE roughly doubles the perplexity cost (+11.5% vs. +5.6% Mistral; +10.1% vs. +5.3% Qwen).
- **Oja baselines**: storing coefficients in the arrival-time basis but reconstructing with the final one gave errors above 1.0. Fixed, OjaKV ties TKV within 0.5%.

When Does TKV-S Work? Measure It

Where do needle tokens fall in TKV's residual distribution?

	TKV residual percentile	needle kept vs. chance	TKV-S vs. SnapKV
TinyLlama	0.66	1.7×	**better**
Mistral-7B	0.64	1.5×	tie
Qwen2.5-14B	0.56	1.1×	worse

The ordering matches exactly. Separability is a **cheap per-model check** on a calibration prompt. Centering the basis did not change Qwen's.

Inside the TKV Cache: TKV-R

In TKV the residual is free and the guarantee applies exactly: keep the top-residual compressed tokens exact instead of evicting them.

- A retained token no longer pays for coefficients it doesn't need: at the same memory that is +33% exact tokens, and 0.83 → **0.93** on Mistral-7B.
- Ranking by the *full* two-term bound picks the same tokens as the key residual: the bound is monotone in $\max \rho^k$, so the simple rule is bound-optimal.
- Same R, W, M (TinyLlama): residual selection **0.67**, SnapKV scores 0.43, H2O scores 0.40.

Mistral-7B, 0.59× memory	PPL	needle
StreamingLLM	**+1.0%**	0.60
TKV, storage only (r64)	+5.6%	0.87
TKV + larger window	+1.9%	0.40
TKV-R r16	+1.8%	0.73
TKV-R r32	+2.5%	**0.93**

Less than half the perplexity cost of plain TKV, at higher retrieval (0.93 vs. 0.87). (Eviction still wins on perplexity alone: TKV's basis is worth more as a *signal* than as storage.)

Limitations

- One synthetic needle format, 2K–4K context, 30 prompts per cell. RULER / LongBench and 32K+ still needed.
- Weaker than SnapKV on Qwen-14B; no KVzip or quantization baseline.
- Needles inside the 512-token warmup get low residuals (worse at larger r).
- TKV-R's rank/window split is chosen on a **disjoint dev set** (30 prompts, shifted haystack windows), not on the reported prompts.